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Mizan: An Innovative Tool for Building Arabic Morphological Pattern Dictionaries and Enhancing Lexical Resource Development

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Mizan is a computational tool designed to enhance the development of Arabic morphological pattern dictionaries and facilitate the creation of advanced lexical resources for linguistic research. By systematically identifying and classifying morphological patterns (awzān), this tool automates the extraction of words sharing the same patterns, and provides an effective solution to overcome the complex challenges of Arabic morphology. This innovation extends beyond the standard pattern-based analysis to be able to process extensive Arabic corpora in a time-efficient manner, and handle datasets larger than five hundred million words with high precision.

Addressing a critical gap in Arabic computational linguistics, the system introduces a novel approach to extract and analyze different morphological patterns. Unlike traditional manual methods or partial solutions available for linguistic research, *Mizan* offers detailed morphological statistics and promotes the development of comprehensive pattern-based dictionaries to achieve high accuracy. Using precision and recall metrics, empirical evaluations confirm its reliability and effectiveness in recognizing complex linguistic phenomena, including non-standard word forms and rare morphological patterns.

Thanks to its adaptability across various Arabic text genres, involving literary, scientific, and news corpora, this approach proposes a practical framework for linguistic studies. By leveraging big data techniques, it contributes to the development of machine learning applications, artificial intelligence, and annotated linguistic resources. By virtue of its ability to process large-scale datasets through morphological analysis, the system yields valuable insights into computational linguistics, lexicography, and Arabic language education to support ongoing research in these fields.

CCS Concepts: • **Computing methodologies** → **Natural language processing**; • **Applied computing** → *Document management and text processing*; • **Information systems** → *Information retrieval;*

Additional Key Words and Phrases: Morphological pattern analysis, arabic morphology, arabic NLP, large-scale corpus processing, morphological pattern dictionaries, computational lexicography, lexical resource development, awzān extraction

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1 Introduction

The Arabic language displays a rich morphological structure based on a complex root and pattern system. This linguistic complexity presents unique challenges in computational linguistics and **natural language processing** (**NLP**). Unlike Indo-European languages, Arabic relies heavily on derivational morphology, where words are formed by applying patterns (أوزان, awzān) to roots, generating a vast array of lexical forms [6, 15, 47]. Although this system contributes to the expressive power of languages, it also complicates tasks such as text analysis, machine translation, and lexical resource development. Despite the remarkable progress made in NLP, Arabic still lacks sufficient computational resources, tools, and annotated corpora, especially for morphological pattern extraction [5, 10, 38].

A key resource in addressing these challenges is morphological pattern dictionaries (صرفية معاجم الأبنية ال), which systematically document both standard (قياسي) and non-standard (سماعي) morphological patterns. Standard patterns are productive and follow regular derivational rules, such as *fa'ala* (فَعَلَ), *maf'ūl* (مَفْعُول), and *istif'āl* (اِسْتِفْعَال), and are commonly used to form new lexical items. In contrast, non-standard patterns are often irregular or unproductive, such as *fu'ay'il* (فُعَيْعِل), and *fu'aylil* (فُعَيْلِل), and are typically confined to a limited set of lexical items preserved in traditional dictionaries rather than generated by rule.

These dictionaries play a vital role in linguistic studies, and help researchers understand Arabic morphology as well as facilitate the development of NLP applications. Traditionally, their compilation has relied on manual efforts, which are labor-intensive, prone to inconsistencies, and unsuitable for large-scale linguistic analysis. This gap highlights the need for automated tools capable of efficiently extracting and classifying Arabic morphological patterns while maintaining high accuracy and scalability [31, 39].

Despite their crucial role in linguistic analysis, morphological pattern dictionaries have historically been scarce, primarily due to the complexity of Arabic morphology and the challenges associated with data organization. Early efforts to document Arabic morphology were embedded in broader linguistic works, such as *Sībawayh's* (d. 180 AH/796 CE) *Al-Kitāb* and *Abū 'Ubayd's* (d. 224 AH/838 CE) *Al-Gharīb al-Muṣannaf.*

These texts, while invaluable, offered scattering morphological insights rather than structured and dedicated compilations [3, 30]. Additionally, historical sources mention other specialized works on morphological patterns, such as *Kitāb al-Abniyah* by *Abū Zakariyyā al-Farrā'* (d. 207 AH/822 CE) and *Kitāb al-Awzān* by *Abū al-hasan al-azdī*, known as *Karrā' al-Naml* (d. 310 AH/922 CE) [31].

However, these works have not survived, and their manuscripts remain undiscovered, leaving their content and methodologies largely unknown. This loss underscores the challenges in tracing the evolution of Arabic morphological studies [31].

Over time, more specialized dictionaries have emerged, focusing exclusively either on noun patterns, like *Al-Maqsūr wa-l-Mamdūd* by *Al-Qālī* (d. 356 AH/967 CE), or verb patterns, such as *Fa'alta wa-Af'alta* by *Al-Zajjāj* (d. 311 AH/923 CE). The development of comprehensive morphological dictionaries is one of the most significant advancements in the history of the Arabic language. In this vein, some examples can be illustrated, namely *Al-Fārābī's* (d. 339 AH/950 CE) *Dīwān al-Adab* that set the standard for this genre, followed by *Nashwān al-Himyarī's* (d. 573 AH/1178 CE) *Shams al-'Ulūm*, that adopted Al-Fārābī's approach in classifying and organizing patterns. Though *Al-Zamakhsharī's* (d. 538 AH/1134 CE) *Muqaddimat al-Adab* lacks the systematic organization of *Dīwān al-Adab*, it is still considered a notable work [25–29].

Beyond these three comprehensive works, most Arabic linguistic efforts did not compile morphological patterns in stand-alone dictionaries but instead integrated them into broader lexicographical

or thematic dictionaries, such as *Ibn Sīda's* (d. 458 AH/1066 CE) Al-Mukhaṣṣas that devoted a section to morphological patterns, or encyclopedic lexicons, like *Al-Suyūtī* (d. 911 AH/1505 CE) *Al-Muzhir* [1, 2]. These early works laid the foundation for Arabic morphological analysis, yet they remained limited in scope and methodology. Reliance on manual compilation struggled to keep pace with the vast and evolving nature of the Arabic lexicon. Consequently, modern computational techniques have become essential to overcome these constraints.

Given the historical challenges and limitations of manual methods, a robust computational approach is required to automate the extraction and classification of Arabic morphological patterns. To bridge this gap, we introduce *Mizan* as a computational tool specifically developed to construct comprehensive morphological pattern dictionaries, enhance lexical resource development, and facilitate large-scale linguistic analysis.

This tool automates the analysis and extraction of morphological patterns from large text corpora and offers a structured and scalable solution for Arabic morphological processing. Unlike traditional systems that used root-based or stem-based analysis [14, 34, 41], *Mizan* focuses on pattern-based extraction, by enabling researchers to systematically identify, classify, and analyze words based on their morphological templates. This approach not only facilitates the creation of comprehensive dictionaries but also provides valuable insights into the frequency distribution of morphological patterns across diverse text genres.

The primary objectives are threefold:

- Automated Morphological Pattern Analysis: Extracting and systematically classifying words based on their morphological structures to improve linguistic precision.
- Scalability and Processing Large Corpora: Ensuring efficient processing of various Arabic text corpora, including literary, scientific, and journalistic texts.
- Lexical Resource Optimization: Enhancing the quality of morphological dictionaries, linguistic databases, and NLP applications based on detailed morphological insights and structured pattern classification.

The complexity of the morphological system in Arabic stems from its users' ability to generate multiple forms from a single root word. Each form carries distinct grammatical functions and meanings. For instance, the root "ف-ل-ك" (f-l-k) can be used to produce diverse patterns, some of which exhibit structural shifts that challenge the conventional morphological analysis. These variations highlight the need for precise and systematic approaches to study Arabic morphology, one of the key objectives of *Mizan* [18, 20, 24].

To this end, Table 1 presents a set of morphological patterns derived from the aforementioned root.

By addressing these challenges, the tool bridges the gap between traditional linguistic resources and modern computational methodologies. For example, it transforms root-and-pattern-based lexical data into structured, machine-readable formats, facilitating integration with NLP pipelines [11, 46]. It also enables automated extraction of morphological paradigms and frequency-based analyses across large corpora, which traditionally required extensive manual effort [22, 45]. These capabilities allow the tool to support tasks such as lexicon expansion, morphological disambiguation, and pattern-based POS tagging, aligning with recent advances in Arabic NLP [17].

Furthermore, its adaptability ensures relevance across various domains, including machine learning, artificial intelligence, and language education.

This article displays the design, implementation, and evaluation of the system and highlights its major contributions to Arabic computational linguistics. Empirical evaluations showcase its effectiveness in extracting morphological patterns, scalability in handling large datasets, and high potential to enhance the development of lexical resources. By integrating this approach into broader

Table 1. Morphological Variations Derived from the Root ف-ل-ك (f-l-k)

No.	Form	Pattern	Pronunciation	Grammatical Function	Meaning	Example
1	فَلَكَ	فَعَلَ (fa-‘a-la)	Fa-la-ka	Verb (past tense)	He twisted or coiled	فَلَكَ الحبلَ بإحكام (He twisted the rope tightly.)
2	فَلَّكَ	فَعَّلَ (fa‘‘a-la)	Fal-la-ka	Verb (intensive form)	He rounded something	فَلَّكَ الخزّاف الطّين ليُشكّل وعاءً (The potter rounded the clay to shape a vessel.)
3	فُلْك	فُعْل (fu‘l)	Ful-k	Noun	Ship	ركبَ نوح الفُلْك (Noah boarded the ark.)
4	فَلَك	فَعَل (fa‘al)	Fa-lak	Noun	Orbit or celestial sphere	الأرض تدور في فَلَك محدّد (The Earth revolves in a fixed orbit.)
5	ف + لك	—	Fa + lak	Particle + Pronoun	So, for you	فَلَكَ أن تختار (So, it’s for you to choose.)

linguistic studies and computational frameworks, researchers can gain deeper insights into Arabic morphology, support the creation of automated dictionary-building tools, and advance AI-driven NLP applications. This system introduces several innovations that enhance the morphological analysis of Arabic:

- `Specialized Pattern Extraction:` Designed for root analysis, frequency studies, and lexicon development, this tool prioritizes pattern recognition over full parsing.
- `Mathematical Model-Driven Analysis:` Unlike lexicon-dependent methods, the approach employs mathematical modeling (e.g., equations and algorithms) to identify morphological structures and enhance scalability.
- `Flexible Pattern Recognition:` Users can define and apply custom templates using an advanced regular-expression engine that will be suitable for various linguistic research needs.
- `Dynamic Affix Handling:` The tool systematically applies rules to remove common prefixes and suffixes, reduce false positives, and improve precision.
- `Adaptive Stop Word Filtering:` Its filtering system intelligently excludes functional words (e.g., particles and conjunctions) based on a set of customizable rules, and ensures meaningful pattern extraction.
- `Scalable Processing:` Designed for large corpora, the framework leverages multi-threading and chunk-based processing to achieve high efficiency.
- `Integrated Statistical Analysis:` Built-in tools generate frequency distributions, unique word counts, morphological summaries, and output options in both visual and tabular formats.

The remainder of this article is structured as follows: Section 2 provides a detailed review of existing studies and tools by identifying the major gaps in contemporary methodologies. Section 3 elaborates on the mathematical framework, design principles, and the computational implementation of the proposed system, offering insights into its algorithmic structure and linguistic modeling. Section 4 assesses the efficiency, scalability, and accuracy of the system using a large-scale Arabic corpus that incorporates detailed performance metrics. Section 5 examines the practical applications of the system in lexicon development, NLP pipelines, and linguistic research, alongside a discussion of computational challenges and potential enhancements. Section 6 presents key findings and

analyses morphological trends, system performance, and linguistic implications derived from the evaluation phase. Section 7 summarizes the study's main contributions, future directions, and broader significance in enhancing Arabic computational linguistics.

2 Related Work

2.1 An Overview of Arabic Morphological Analysis Tools

Arabic morphological analysis has significantly evolved due to this morphologically complex language, root-pattern system, and context-sensitive word structures. Various tools using different methods have been developed to overcome these challenges. This section reviews key Arabic morphological analysis tools and their methods, as well as highlights their strengths and limitations.

— **`Buckwalter Arabic Morphological Analyzer (BAMA)`**

Developed by Tim Buckwalter (2004), BAMA is a lexicon-based tool that generates all possible analyses for a given word by combining roots, prefixes, and suffixes. Although it is effective in listing all potential forms, this analyzer lacks disambiguation capabilities and often overgenerates due to its reliance on lexical matching. Despite these limitations, BAMA sets the foundation for later tools by showing the value of structured lexicons and morphological rules [12].

— **`Standard Arabic Morphological Analyzer (SAMA)`**

An enhanced version of BAMA developed by the Linguistic Data Consortium, SAMA features a larger lexicon and presents morphological rules. It extends support for numerous Arabic dialects by increasing its applicability to enhance both research and commercial applications. However, like the former analyzer, SAMA depends heavily on lexicons and requires continuous updates to accommodate evolving word forms [36].

— **`Morphological Analysis and Disambiguation for Arabic (MADA)`**

Introduced by Habash and Rambow [23], MADA builds on BAMA by incorporating machine-learning techniques to resolve morphological ambiguities. It integrates morphological analysis, part-of-speech tagging, and disambiguation to significantly improve performance in large-scale Arabic NLP applications. However, MADA relies on extensive annotated corpora for training and can be computationally intensive.

— `AMIRA`

Developed by Mona Diab, AMIRA combines linguistic rules with statistical modeling for tokenization, part-of-speech tagging, and base phrase chunking. Optimized for **Modern Standard Arabic** (**MSA**), it prioritizes efficiency and it is suitable for various NLP applications. However, its focus on formal Arabic limits its effectiveness for dialectal variations [16].

— `MADAMIRA`

MADAMIRA merges features from both MADA and AMIRA, offering a comprehensive tool for tokenization, diacritization, and morphological analysis. It employs a hybrid approach that balances linguistic rules and statistical models, improving both accuracy and efficiency. However, similar to MADA, this analyzer requires significant computational resources for large datasets [41].

— `Farasa`

Developed by the **Qatar Computing Research Institute** (**QCRI**), Farasa is a fast and scalable Arabic morphological analyzer. It employs a hybrid approach that combines rule-based and machine-learning techniques to improve tasks, such as text segmentation, lemmatization, and morphological tagging. Optimized for real-time processing and large-scale NLP tasks, this analyzer's performance may vary depending on a specific text genre type and domain [14].

— `CamelMorph MSA`

More recently, CamelMorph MSA has been introduced as a large-scale, open-source morphological analyzer for Modern Standard Arabic. Developed by the CAMeL-Lab, this tool extends previous efforts to construct a broad-coverage lexicon and enhance handling **out-of-vocabulary** (**OOV**) words. It uses lexicon-based analysis to reduce reliance on annotated training data while maintaining competitive accuracy. However, its performance with dialectal variations and irregular morphological structures needs further investigation [33, 34].

All these tools have significantly advanced in Arabic NLP. While early tools like BAMA established foundational methodologies, later developments such as MADA, SAMA, MADAMIRA, and CamelMorph MSA have integrated advanced algorithms and lexicon-based strategies to enhance accuracy and scalability. However, most of these tools prioritize full morphological parsing, which may not always align with research objectives that emphasize pattern extraction or frequency analysis.

2.2 Research Gaps and Limitations of Existing Tools

Despite notable advancements, existing Arabic morphological analysis tools face several key limitations that hinder their efficiency and adaptability:

— `Dependence on Lexical Databases:` Many tools, such as BAMA [12] and SAMA [36], rely heavily on predefined lexicons that require constant updates and may not cover new word forms.
— `Limited Scalability:` Rule-based systems often struggle with processing large corpora efficiently, leading to computational overhead.
— `Lack of Adaptability:` Although they allow retraining, machine learning-based tools demand extensive annotated datasets and significant computational resources.
— `Handling Non-Standard Forms:` Most tools are optimized for Modern Standard Arabic (MSA). However, they perform poorly on dialectal variations.
— `Customization Challenges:` Many systems offer limited flexibility, making adaptation for specialized complex linguistic research.

Addressing these limitations requires a more adaptive approach that balances linguistic accuracy and computational efficiency. The next section examines how existing methodologies are compared in terms of scalability, adaptability, and linguistic precision, paving the way for a more effective framework for Arabic morphological analysis.

2.3 Addressing Limitations in Arabic Morphological Analysis

While both traditional and modern Arabic morphological analysis tools have significantly contributed to this field, they exhibit notable limitations that affect their effectiveness in large-scale linguistic research. Traditional tools primarily rely on rule-based methods and extensive lexicons, making them rigid and difficult to expand. On the other hand, modern tools leverage hybrid machine learning approaches, based on large annotated corpora and computationally expensive training. However, they may lack adaptability to specialized linguistic research needs.

Table 2 provides a comparative overview of traditional and modern approaches for Arabic morphological analysis, highlighting their methodological differences and inherent limitations.

While modern machine learning approaches enhance scalability and adaptability, they remain computationally demanding and require substantial annotated resources. The next section introduces an optimized framework that balances computational efficiency and linguistic accuracy for Arabic morphological analysis.

Table 2. A Comparative Overview of Traditional and Modern Arabic Morphological Analysis Approaches

Aspect	Traditional Approaches (Rule-Based and Lexicon-Dependent) - (e.g., BAMA, SAMA)	Modern Approaches (Machine Learning and Hybrid) - (e.g., MADAMIRA, Farasa, CamelMorph)
Methodology	Applies pre-defined morphological rules and structured lexicons.	Uses statistical models and machine learning techniques for pattern recognition.
Database Dependency	Requires extensive lexicons with manual updates.	Trained on large annotated corpora, reducing reliance on pre-defined lexicons.
Customization	Limited flexibility, requires manual rule adjustments.	Allows retraining and adaptation to new linguistic data.
Scope of Analysis	Focuses on comprehensive morphological parsing.	Prioritizes token-based analysis and probabilistic models.
Scalability	Struggles with processing large datasets due to computational overhead.	More scalable but computationally intensive, requiring high processing power.
Handling of Affixes	Relies on static affix lists, which limit adaptability.	Learns affixation patterns extracted from training data, allowing for greater flexibility.
Stop Word Filtering	Uses pre-defined stop word based on a well-defined manual curation process.	Applies contextual models to determine stop words dynamically.
Applications	Mainly used in linguistic research and parsing tasks.	Supports NLP applications, like machine translation and sentiment analysis.

3 Methodology and Mathematical Framework

This section outlines the methodological and mathematical foundations of Mizan, a rule-based tool designed for large-scale morphological pattern analysis in Arabic corpora. The core concept is centered on the use of user-defined morphological templates—canonical patterns that represent root-based morphological structures in Arabic (e.g., *istifʿāl* (استفعال), *tafaʿʿala* (تفعّل)). These templates are transformed into regular expressions and applied across large corpora to detect word forms that match the specified patterns across diverse genres and styles. The system incorporates affix management, stopword filtering, and multithreaded processing to ensure scalability and linguistic precision. The extracted data is then structured into lexical resources, including frequency lists, root-pattern associations, and part-of-speech tags. The following subsections detail the internal design, pattern processing techniques, formal mathematical modeling, and performance considerations of the system.

3.1 A Technical Overview of the Tool: Design, Structure, and Algorithms

Mizan represents a specialized computational framework designed for large-scale processing and analysis of Arabic text corpora. It integrates linguistic principles with algorithmic precision to extract and classify morphological patterns (awzān), facilitates the development of structured morphological dictionaries and supports NLP applications.

3.1.1 Design Principles and Structure.

— `Corpus Preparation`
To ensure optimal performance and accuracy, *Mizan* preprocesses input corpora following these steps:
- Text Cleaning and Normalization: Removing extraneous characters and standardizing diacritics where applicable.
- Segmentation for Efficient Processing: Breaking large text files into smaller chunks to optimize memory use and execution time.

  - Preliminary Statistical Analysis: Computing key corpus metrics (e.g., total word count, unique token count) as a foundational baseline for subsequent analysis.

  These preprocessing steps align with the best NLP practices as outlined by [37] and statistical NLP methods as shown in a study conducted by [32].

— `Morphological Pattern Matching` *Mizan* employs user-defined morphological templates—root-based structures such as CVCVC, *istifʿāl* (استفعال), and *tafaʿʿul* (تفعل) that are converted into regular expressions (regex) for efficient pattern detection across the corpus.
  - Prefix and Suffix Handling: *Mizan* dynamically integrates predefined lists of prefixes (e.g., ال, بال) and suffixes (e.g.,ها, ـه) to enhance pattern recognition and account for derived forms.
  - Stop Word Filtering: High-frequency function words (e.g., إذ, إذا) are excluded to maintain focus on meaningful lexical items.

  By leveraging finite-state morphology, as described by [9], the system ensures efficient affix handling and pattern recognition. Additionally, ambiguity resolution strategies proposed by [7] contribute to the accurate identification of overlapping morphological structures. Furthermore, theoretical perspectives suggested by [21] and the Distributed Morphology framework [42] provide a conceptual foundation for understanding how morphological patterns are decomposed into structured representations and can be computationally modeled. For instance, the *istifʿāl* (استفعال) template is encoded using the regex pattern /^ا..است.*/, allowing for a precise identification of words that confirm this structure while filtering out irrelevant forms.

— `Dictionary Integration and Lexical Resource Development`
  Building upon its pattern recognition capabilities, *Mizan* is integrated with the existing lexicons such as "*al-Muʿjam al-Wasīt*" to facilitate systematic lexicon expansion. By aligning newly extracted morphological data with entries from such authoritative dictionaries, *Mizan* enables the structured enhancement and extension of traditional lexical content, particularly in root-based and pattern-oriented organization. This process involves:
  - Lexical Database Enhancement: Results are stored in structured formats (e.g., CSV, spreadsheets) for seamless integration of external lexical resources.
  - Pattern-based Dictionary Construction: The extracted words are categorized into morphological paradigms to help construct dictionaries that capture Arabic word formation rules with high accuracy.

  By structuring extracted data in a systematic manner, *Mizan* not only analyzes text corpora but also generates reliable resources for lexicography and NLP applications.

3.1.2 *Algorithmic Framework.*

— `Parallel Processing for Large-scale Analysis`
  To ensure scalability and efficiency, *Mizan* makes use of parallel processing:
  - Segment-based Processing: Large corpora are split into smaller units to be processed concurrently, reducing execution time while preserving accuracy.
  - Scalability for Large Text Collections: This approach allows the analysis of corpora that exceed 500 million tokens.

— `Statistical Aggregation and Reporting`
  For each morphological pattern, *Mizan* computes:
  - Total occurrences of words matching the pattern.
  - Unique terms adhering to the pattern.
  - Frequency distributions and corpus coverage percentages.

The extracted data is exported in structured formats to support further analysis and visualization, and ensure adaptability across various linguistic research fields.

— `Enhancements and Adaptability Features`
Mizan introduces several innovations to improve its adaptability and effectiveness in linguistic research:
  - Customizable Templates: Researchers can define or modify regex templates to align with specific linguistic hypotheses or research objectives.
  - Handling Non-Standard Morphological Structures: *Mizan* incorporates ambiguity resolution techniques, inspired by [7], in order to enhance its ability to recognize rare and non-standard word forms.
  - Comprehensive Outputs: Generated reports include token breakdowns, pattern frequency analysis, and morphological variations, supporting both exploratory and confirmatory linguistic research.

By integrating robust morphological analysis with large-scale corpus processing, this framework significantly improves different fields, such as Arabic computational linguistics, lexicography, as well as NLP research, and ensures more accurate and efficient linguistic analysis.

3.2 Morphological Template Processing and Affix Management

The effectiveness of the system depends on two fundamental components that ensure accurate extraction of morphological patterns in Arabic text corpora.

3.2.1 Morphological Template Processing: Users can define morphological templates (e.g., *tafaʿʿul* (تفعُّل), *istifʿāl* (استفعال)) as structured patterns. These templates undergo validation to ensure they consist exclusively of Arabic letters and comply with predefined syntactic constraints. Once validated, the tool applies these templates to the corpus by systematically extracting matching words and analyzing their distributions.

The implementation follows a structured decision workflow, establishing consistent and efficient processing.

The validation stage aims at not only ensuring that the template conforms to Arabic morphological rules, but also checking structural correctness (e.g., valid root-based formations). Next, pattern matching helps convert the template into a regular expression (regex) for efficient text processing, and then scan the corpus to identify words conforming to a specific pattern. In the last phase, known as extraction and storage, users extract all matching words and store the results in structured formats (e.g., CSV, spreadsheets). Finally, frequency statistics are computed to support large-scale corpus analysis. This workflow is visually represented in the following decision tree (Figure 1).

3.2.2 Affix Handling and Stop Word Filtering. To enhance precision in pattern extraction, *Mizan* integrates curated linguistic resources, including affix lists and stop word filters.

Concerning Prefix and Suffix Management, *Mizan* dynamically applies pre-defined lists of common prefixes (e.g., بال, ال) and suffixes (e.g., ـه,ـها) to improve the recognition of derived forms. This approach accounts for common morphological variations and ensures a more comprehensive analysis of word structures by leveraging pattern-based generalizations. Unlike conventional analyzers such as BAMA and Farasa, which rely on static lexicons or corpus-trained models, Mizan's method offers greater adaptability and transparency.

As an illustration of Mizan's output, consider the word "*istikhdām*" (استخدام=employment), which appears under the pattern *istifʿāl* (استفعال) in Table 6.

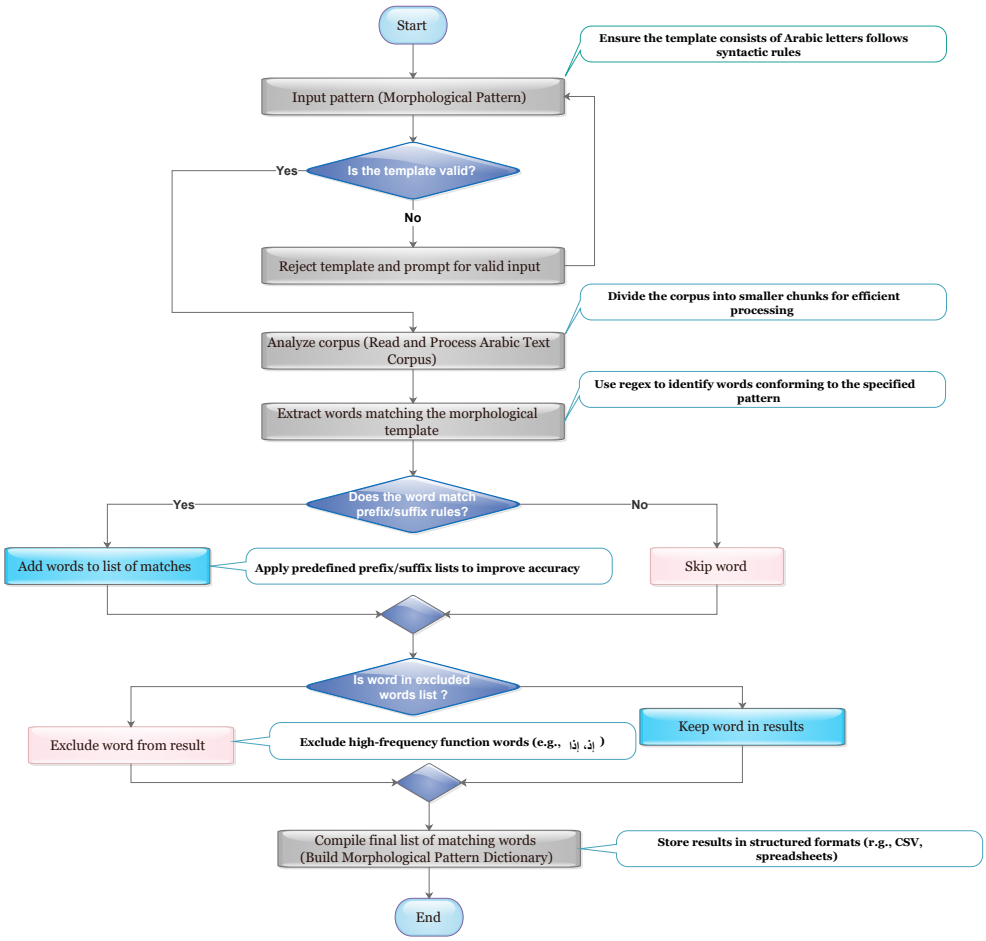

Fig. 1. Decision tree for validating and analyzing morphological patterns in Arabic corpora.

The tool analyzes this word as a verbal noun (maṣdar) with the pattern *istifʿāl* (استفعال), extracting the root "خ-د-م". It identifies the nominal status of the form and classifies it within the paradigm of derived verbal nouns. This analysis is produced automatically through pattern-based recognition, without reliance on predefined lexical entries, demonstrating Mizan's ability to accurately deconstruct and categorize morphologically derived forms.

As for Stop word Filtering, high-frequency function words (e.g., إذ, إذا) are excluded to maintain focus on semantically meaningful items as previously noted in Section 3.1.1. This is implemented here through integrated stopword lists, which refine the extraction process and reduce noise. This step ensures that extracted words primarily contribute to morphological pattern research rather than being influenced by frequent and non-informative words.

By incorporating these components, *Mizan* achieves higher accuracy and scalability, and enhances a comprehensive and computationally efficient morphological analysis process. These strategies align with established methodologies in Arabic computational linguistics and further enhance the tool's applicability to large-scale corpora.

3.3 A Formal Mathematical Model for Arabic Morphological Analysis

This computational framework utilizes a structured mathematical approach to analyze Arabic morphological patterns, ensuring precision and scalability in handling vast text corpora. Below are the key components formalized mathematically.

3.3.1 Corpus Representation and Input Modeling. Corpus C consists of N textual documents, where each document d_i contains sentences S_j, and each sentence includes different words w_k:

$$C = \{d_1, d_2, \dots, d_N\},\ d_i = \{S_1, S_2, \dots, S_K\},\ Sj = \{w_1, w_2, \dots, w_M\}$$

Each word w_k satisfies $\ni w_k$ an Arabic script, $\forall k$, constructing simple processing unit.

3.3.2 Template Definition. Morphological templates T are based on classical Arabic patterns, like *fa'ala* (فعل). A word w_i matches a template T if it follows the pattern:

$$T = \text{“فعل”} \Rightarrow w_i = \text{“فَعَلَ”} \text{ where } w_i \in C$$

Templates can be expanded by adding prefixes P and suffixes S:

$$T_{expanded} = P.T.S$$

where "·" denotes string concatenation.

3.3.3 Pattern Matching. A regular expression R is generated based on the expanded template:

$$R = Regex(P.T.S)$$

where P and S account for common affixes while T maintains the core morphological structure.

A word w is considered as a match for a pattern if and only if:

$$w \in C \text{ and } w \in R \Rightarrow w \in T_{matched}$$

leading to strict pattern conformance. To prevent false positives, words that deviate from the expected template beyond a predefined threshold (e.g., in vowel shifts or inserted letters) are excluded based on morphological constraints.

3.3.4 Affix Handling. To isolate the root form of a word, affixes are removed using the following function:

$$w' = Trim(w, S, L, \lambda)$$

where S denotes prefixes, L represents suffixes, and λ models internal morphological shifts, such as كُتِبَ → كَتَبَ where the vowel shift ('a' → 'u', 'a'→ 'i') is accounted for.

This step ensures a robust stem extraction algorithm even in cases of morphophonemic alterations. However, to further improve accuracy, *Mizan* uses exception handling for irregular verb forms and other non-standard derivations. For instance, verbs exhibiting internal vowel shifts (e.g., يقول → قال) are processed using a set of pre-defined transformation rules that account for phonological alternations beyond regular affix removal systems.

3.3.5 Stop Word Filtering. Common function words are excluded, as shown in the following filter function:

$$Filter(w_i) = \begin{cases} w_i, & \text{if } w_i \notin \text{StopWords} \\ \emptyset, & \text{if } w_i \in \text{StopWords} \end{cases}$$

3.3.6 Statistical Analysis. For each matching word w, its frequency is computed:

$$\text{freq}(w) = \sum_{d_i} \text{count}(w, d_i)$$

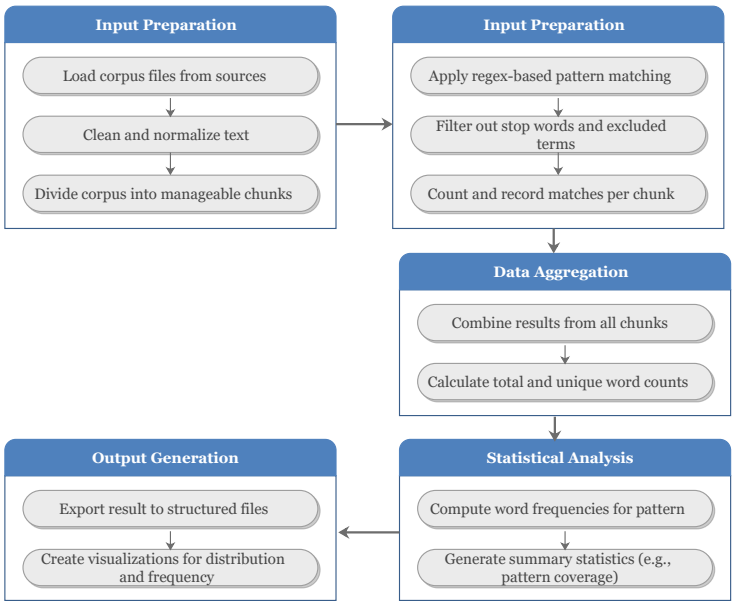

Fig. 2. A flowchart of Arabic corpora processing and morphological pattern extraction for lexical resource development.

The total number of unique words U and the number of word occurrences F are counted, as presented in the following equation:

$$U = |T_{matched}|, F = \sum_{w \in T_{matched}} \text{freq}(w)$$

Additionally, the relative distribution of morphological patterns is analyzed as follows:

$$P(T) = \frac{freq(T)}{F}$$

where $P(T)$ represents the proportion of words conforming to a given template relative to the entire corpus. This metric offers deeper insights into the predominance of specific morphological structures in Arabic texts and helps quantify their linguistic significance. The overall workflow for processing the Arabic corpus and generating statistical outputs based on morphological patterns is depicted in Figure 2. The latter highlights key processing steps, such as affix removal, stop-word filtering, and statistical analysis.

This formalization establishes a rigorous computational framework that systematically extends the methodologies outlined in the previous sections (Sections 3.1 and 3.2) and ensures consistency between theoretical modeling and practical implementation in morphological pattern extraction.

Given the vast size of Arabic corpora and the computational complexity of morphological analysis, efficient processing mechanisms are essential to ensure scalability. The following section outlines how *Mizan* employs multithreading to parallelize the analysis process, optimizing both speed and resource utilization while maintaining the desired extraction accuracy.

3.4 Efficient Processing of Large-scale Textual Data Using Multithreading

To efficiently scale the morphological analysis process described in Section 3.3, *Mizan* employs a multithreading approach designed for large-scale text processing. By distributing computational

load across multiple threads, the system optimizes both performance and resource utilization while ensuring robust morphological analysis.

3.4.1 Chunk-based Distribution. Given a corpus C, where $|C| >> 10^6$ words, the corpus is broken down into smaller chunks C_i while preserving morphological boundaries to enable parallel processing without disrupting pattern recognition.

$$\bigcup C_i, \forall i, |C_i < chunksize$$

This ensures that each processing thread handles a manageable piece of data and reduces memory bottlenecks.

3.4.2 Parallel Execution. Each worker thread processes a chunk C_i using the following function:

$$Thread_i = ProcessChunk(C_i)$$

This reveals that all identified morphological patterns are consolidated efficiently, while preserving uniqueness and maintaining data integrity for final statistical reporting purposes.

3.4.3 Time Complexity Analysis. Time complexity of the processing model is reduced by parallel execution, as demonstrated in the following equation:

$$T_{complexity} = O\left(\frac{freq(N)}{T}\right)$$

where N is the total number of words in the corpus, and T is the number of threads used. This assumes uniform task distribution and minimal synchronization delays. However, the load imbalance between threads may introduce slight variations in processing efficiency, particularly when handling highly inflected words.

Figure 3 illustrates the multithreading approach in *Mizan*, detailed task assignment, parallel processing, synchronization, and result aggregation.

4 Evaluation

4.1 Design of the Experiment

The major purpose of this evaluation is to measure *Mizan*'s effectiveness in extracting morphological patterns accurately and processing large-scale corpora with optimal efficiency [4]. The Arabic Wikipedia corpus (arwiki-20241120) was selected for the experiment due to its different characteristics:

- Open-Source Accessibility: Freely available under the GNU Free Documentation License.
- Scale: Over 572 million words (572,772,346 words) with 2,474,755 unique word forms, providing a substantial dataset for testing scalability.
- Diversity: Covers a wide range of topics and diverse writing styles, reflecting the richness of Arabic morphology.
- Linguistic Morphological Complexity: Its extensive morphological patterns make it an optimal testbed for different linguistic analysis tools.

To ensure comprehensive assessment, six distinct morphological patterns were chosen depending on their structural complexity as follows:

- Four-character patterns: *fa'ūl* (فعول), *fa'īl* (فعيل) [e.g., (*karīm* (كريم = "generous")), (*ṣabūr* (صبور = "patient"))]
- Five-character patterns: *fā'ūl* (فاعول), *maf'ūl* (مفعول) [e.g., (*maktūb* (مكتوب = "written")), (*nāqūs* (ناقوس = "bell"))]
- Six-character pattern: *mutafā'il* (متفاعل) [e.g., (*muta'āwin* (متعاون = "cooperative"))]
- Seven-character pattern: *istif'āl* (استفعال) [e.g., *istighfār* (استغفار = "seeking forgiveness")]

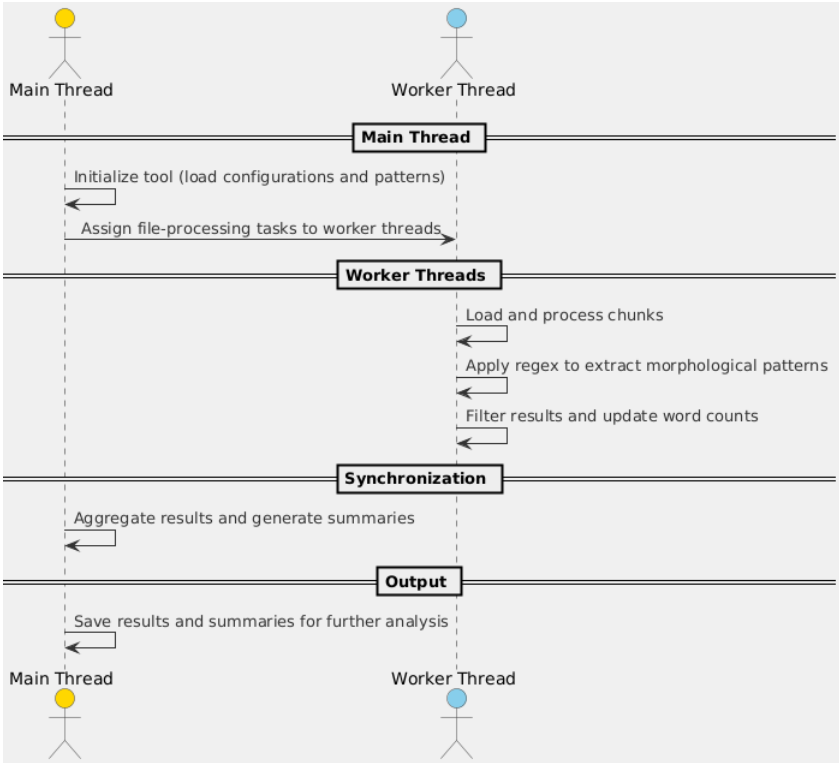

Fig. 3. Optimized multithreading in *Mizan* for scalable processing of morphological patterns and lexical resources.

Although both verbal and nominal patterns were considered, nominal patterns were prioritized in the evaluation because they are more challenging to identify computationally. Unlike verbal patterns, which often include clear morphological markers indicating tense and voice, nominal patterns lack such consistent surface cues. This makes their automatic extraction a more demanding and revealing test of the system's precision. The selected patterns also span a variety of morphological lengths and derivational mechanisms (including prefixation and infixation), making them well-suited for evaluating the tool's accuracy and coverage.

This evaluation process put special focus on two key aspects:

(1) Accuracy of Morphological Pattern Identification: evaluating *Mizan*'s ability to accurately detect frequently occurring words and identify anomalies such as borrowed or irregular forms that do not conform to standard morphological patterns (e.g., موبايل (Mobile), إنترنت (Internet)).
(2) Computational Efficiency in Large-scale Processing: Measuring *Mizan*'s performance in handling massive corpora while offering meaningful statistical insights.

4.2 Accuracy Metrics

To test accuracy metrics, namely, *Precision*, *Recall*, and $F1-Score$, we selected a standardized sample of the 1,000 most-frequent words for each morphological pattern. This decision was motivated by the need to achieve a balance between representativeness and analytical feasibility. Frequent words tend to reflect stable grammatical and lexical structures shaped by productive morphological patterns, making them useful indicators of pattern-governed linguistic behavior and corpus usage.

Table 3. Overview of Dataset Metrics for Morphological Patterns

Pattern	Total number of Words in the Corpus	Words in the Sample (With Repetition)	Percentage of Sample Outputs compared to the Total Corpus Outputs	Correct Words (Unique)	Unique Words in the Sample
fa'īl (فعيل)	27,238,428	24,213,424	88.91%	803	1,000
fa'ūl (فعول)	13,171,270	11,164,296	84.77%	838	1,000
maf'ūl (مفعول)	2,427,944	2,336,446	96.22%	955	1,000
fā'ūl (فاعول)	1,177,941	1,103,337	93.66%	942	1,000
mutafā'il (متفاعل)	112,746	112,254	99.56%	1,000	1,000
istif'āl (استفعال)	4,633,858	4,618,487	99.67%	1,000	1,000

Table 4. Summary of Precision, Recall, and F1-Score for Various Patterns

Pattern or Math	Precision (%)	Recall (%)	F1-Score (%)
fa'īl (فعيل)	80.3	80.3	80.3
fa'ūl (فعول)	83.8	83.8	83.8
maf'ūl (مفعول)	95.5	95.5	95.5
fā'ūl (فاعول)	94.2	94.2	94.2
mutafā'il (متفاعل)	100	100	100
istif'āl (استفعال)	100	100	100

By selecting the top 1,000 words for each pattern:

(1) Representativeness is ensured by highlighting the most frequent forms that are central to the characteristics of corpus linguistics.
(2) Manageability is attained when the sample is manually reviewed and accurately analyzed within a reasonable timeframe.
(3) Bias reduction is achieved by choosing a standard sample size, minimizing skewed results from either too small or too large samples.

Table 3 displays an overview of the dataset metrics, including the total number of words in the corpus for each pattern, the total number of occurrences of a word in the sample, and the corresponding unique words (both correct and incorrect words).

The evaluation metrics: *Precision*, *Recall*, and $F1 - Score$ were calculated for each pattern using the following formulas as follows:

$$Precision = \frac{TruePositives(TP)}{TruePositives(TP) + FalsePositives(FP)}$$

$$Recall = \frac{TruePositives(TP)}{TruePositives(TP) + FalseNegatives(FN)}$$

$$F1 - Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Table 4 summarizes the outcomes for each pattern.

Table 4 shows the evaluation metrics for six selected morphological patterns. The results indicate that the tool performs very well across all patterns, with perfect scores (100%) for the more complex six- and seven-character patterns (استفعال, متفاعل). The slightly lower scores for the shorter patterns (فعول, فعيل) likely reflect their higher frequency and greater morphological variability in the corpus, which pose additional challenges for accurate identification. Overall, these results demonstrate the tool's strong capability in accurately extracting and analyzing diverse nominal morphological patterns.

Table 5. Processing Time for Morphological Patterns

Pattern	Time (minutes:seconds)	Time (seconds)
fa'īl (فعيل)	15:35	935.38
fa'ūl (فعول)	15:04	904.40
maf'ūl (مفعول)	14:13	853.72
fā'ūl (فاعول)	14:28	868.89
mutafā'il (متفاعل)	14:06	846.90
istif'āl (استفعال)	16:25	985.55

4.3 Efficiency

The efficiency of *Mizan* was measured using three key dimensions, namely, Processing Time, Scalability, and Resource Usage.

4.3.1 Processing Time. The time required to process each morphological pattern was measured as follows (Table 5).

Table 5 presents the processing times for each morphological pattern during evaluation. The results show relatively consistent processing durations across all patterns, with times ranging from approximately 14 to 16 minutes. The slightly longer time for the *istif'āl* (استفعال) pattern may be due to its greater structural complexity and frequency in the corpus. These timings demonstrate the tool's efficiency in handling large-scale morphological analysis while maintaining accuracy.

Key optimizations involve:

- **Parallel Processing:** Tasks are distributed across multiple processes or threads to handle large data volumes concurrently. This significantly improves throughput and reduces overall processing time, especially when analyzing millions of word forms.
- **Caching:** Intermediate results (such as analyzed patterns or filtered word forms) are retained in memory to avoid redundant computation when similar forms reappear in subsequent segments.
- **Incremental Analysis:** The corpus is processed in smaller, consecutive batches rather than loaded in full. This chunk-wise strategy reduces memory footprint and enables the tool to scale efficiently on systems with limited resources.

4.3.2 Scalability. Mizan was tested on a test corpus extracted from Arabic Wikipedia (approximately 572 million words). The obtained findings demonstrated its ability to handle large datasets efficiently, with processing time that increases proportionally to pattern complexity. Techniques, such as data partitioning and distributed processing (using frameworks like Dask and Apache Spark) ensured scalability.

4.3.3 Resource Usage. Memory and CPU consumption were monitored during processing. *Mizan* made use of:

- **Memory Management:** Caching and lazy evaluation to minimize memory usage.
- **Processor Optimization:** Using libraries like NumPy and Cython for computationally intensive tasks.
- **GPU Processing:** Leveraging GPU capabilities for larger datasets.

Figure 4 illustrates *Mizan*'s performance in processing Arabic morphological patterns. It visualizes user-defined input patterns, real-time progress tracking, and execution time for each pattern. Execution statistics provide insights into the efficiency and responsiveness of the program, highlighting its ability to handle sequential multiple patterns while generating dynamic reports and charts.

```
Python 3.13.1 (tags/v3.13.1:0671451, Dec  3 2024, 19:06:28) [MSC v.1942 64 bit (
AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.

==================== RESTART: D:\NLP tools\Mizan.v.1.0.py ====================
Enter the desired pattern (e.g., استفعال, تفعل, مفعل): فعيل
Do you want to add more patterns? (y/n): y
Enter the desired pattern (e.g., استفعال, تفعل, مفعل): فعول
Do you want to add more patterns? (y/n): y
Enter the desired pattern (e.g., استفعال, تفعل, مفعل): مفعول
Do you want to add more patterns? (y/n): y
Enter the desired pattern (e.g., استفعال, تفعل, مفعل): فاعول
Do you want to add more patterns? (y/n): y
Enter the desired pattern (e.g., استفعال, تفعل, مفعل): متفاعل
Do you want to add more patterns? (y/n): y
Enter the desired pattern (e.g., استفعال, تفعل, مفعل): استفعال
Do you want to add more patterns? (y/n): n
Input statistics generated: D:\NLP tools\Mizan outputs\Statistics_Input.txt
Processing pattern: 0/1 |          |%0   :فعيل [00:00<?, ?it/s]Processing patter
n: 1/1 |[redacted]|%100 :فعيل [16:24<00:00, 984.91s/it]Processing pattern: فعيل:
 100%|[redacted]| 1/1 [16:24<00:00, 984.98s/it]
Processing pattern: 0/1 |          |%0   :فعول [00:00<?, ?it/s]Processing patter
n: 1/1 |[redacted]|%100 :فعول [15:51<00:00, 951.16s/it]Processing pattern: فعول:
 100%|[redacted]| 1/1 [15:51<00:00, 951.20s/it]
Processing pattern: 0/1 |          |%0   :مفعول [00:00<?, ?it/s]Processing patte
rn: 1/1 |[redacted]|%100 :مفعول [14:58<00:00, 898.82s/it]Processing pattern: مفع
1/1 |[redacted]|%100 :ول [14:58<00:00, 898.87s/it]
Processing pattern: 0/1 |          |%0   :فاعول [00:00<?, ?it/s]Processing patte
rn: 1/1 |[redacted]|%100 :فاعول [15:10<00:00, 910.65s/it]Processing pattern: فاع
1/1 |[redacted]|%100 :ول [15:10<00:00, 910.69s/it]
Processing pattern: 0/1 |          |%0   :متفاعل [00:00<?, ?it/s]Processing patt
ern: 1/1 |[redacted]|%100 :متفاعل [14:58<00:00, 898.17s/it]Processing pattern: م
1/1 |[redacted]|%100 :تفاعل [14:58<00:00, 898.23s/it]
Processing pattern: 0/1 |          |%0   :استفعال [00:00<?, ?it/s]Processing pat
tern: 1/1 |[redacted]|%100 :استفعال [15:12<00:00, 912.09s/it]Processing pattern:
 1/1 |[redacted]|%100 :استفعال [15:12<00:00, 912.19s/it]
Pattern statistics generated: D:\NLP tools\Mizan outputs\Statistics_Patterns.txt
Dynamic table and chart saved: D:\NLP tools\Mizan outputs\Dynamic_Report.xlsx
```

Fig. 4. Statistical processing patterns and execution time in *Mizan*.

4.4 Linguistic Analysis of Morphological Patterns

Findings reveal a strong correlation between morphological complexity and classification accuracy, with longer and more structured patterns leading to higher precision. Patterns that follow standard morphological rules (canonical patterns) or have longer structures tend to produce more accurate results.

This is evident in patterns such as *istifʿāl* (استفعال) and *mafʿūl* (مفعول), where their well-defined structure enhances contextual clarity and facilitates correct word identification. This is because such patterns constrain the placement of root letters and affixes in predictable ways, reducing ambiguity and making it easier to distinguish between valid and invalid word forms, even in complex textual contexts. For instance, the pattern *istifʿāl* (استفعال) typically starts with the prefix " است" and follows a predictable root alignment, which helps the system eliminate unrelated forms and improves morphological disambiguation.

The column chart in Figure 5 illustrates the frequency distribution of the six selected patterns extracted from the Arabic Wikipedia corpus. Patterns like *faʿīl* (فعيل) and *faʿūl* (فعول) exhibit higher total frequencies due to their prevalence in various linguistic contexts and greater morphological ambiguity, which increases their data representation. In contrast, patterns such as *mutafāʿil* (متفاعل) and *istifʿāl* (استفعال), though having fewer unique words, make meaningful contributions to overall frequencies based on specialized linguistic roles that reflect their specific usage.

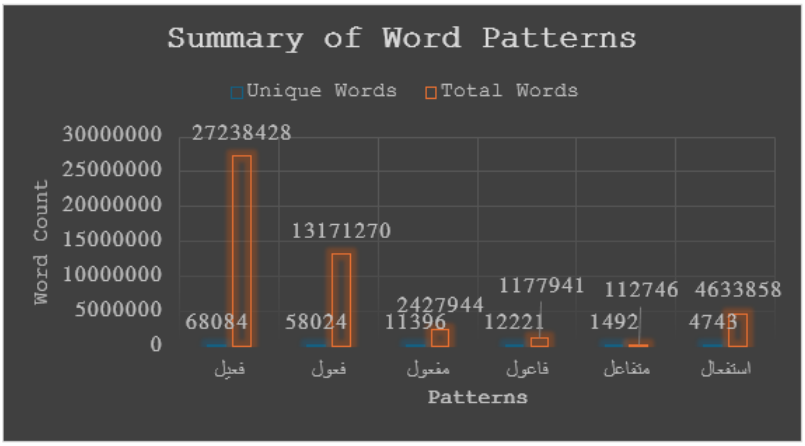

Fig. 5. A column chart showing the frequency of word patterns in Arabic Wikipedia data.

Table 6. Most Frequent Words Across Selected Morphological Patterns

No.	فعيل	Freq	No.	فعول	Freq	No.	مفعول	Freq
1	شريط	1,278,294	1	وجود	173,294	1	مشروع	113,876
2	رئيس	239,420	2	حدود	134,503	2	موضوع	90,103
3	جميع	220,539	3	حقوق	88,301	3	محمود	78,041
4	فريق	161,341	4	حضور	86,858	4	مفهوم	46,077
5	كبير	141,714	5	ركوب	64,081	5	معروف	40,005
No.	**فاعول**	**Freq**	**No.**	**متفاعل**	**Freq**	**No.**	**استفعال**	**Freq**
1	قانون	109,528	1	متكامل	12,682	1	استشهاد	3,454,094
2	حاسوب	20,182	2	متسابق	9,847	2	استخدام	158,300
3	قاموس	15,968	3	متزايد	7,243	3	استبدال	63,349
4	صاروخ	15,839	4	متزامن	2,109	4	استعمال	32,940
5	فاروق	11,049	5	متساوي	1,939	5	استرجاع	32,399

Table 6 lists the most frequent words for each morphological pattern, providing insights into the linguistic behavior of these patterns.

Conversely, short patterns or those considered as non-standard irregular patterns exhibit low performance due to increased ambiguity, as further illustrated in Figure 5.

Additionally, the integration of loanwords into the Arabic lexicon plays a vital role in the observed discrepancies. Shorter morphological patterns appear more susceptible to hosting foreign words, reducing their precision in strictly identifying native Arabic terms.

These findings underscore the crucial role of morphological complexity and standardization in achieving accurate linguistic analysis, while also revealing the influence of external linguistic factors, such as loanwords, which are words borrowed from other languages and integrated into Arabic (e.g., “راديو” (radio), “كمبيوتر” (computer]) on classification accuracy, as such words may not conform to traditional Arabic morphological patterns. Moreover, shorter patterns, such as “فَعَلَ” or “فِعْل” (e.g., علم, عِلم, كتب), tend to be more ambiguous and morphologically dense, presenting additional challenges in accurate pattern-based classification.

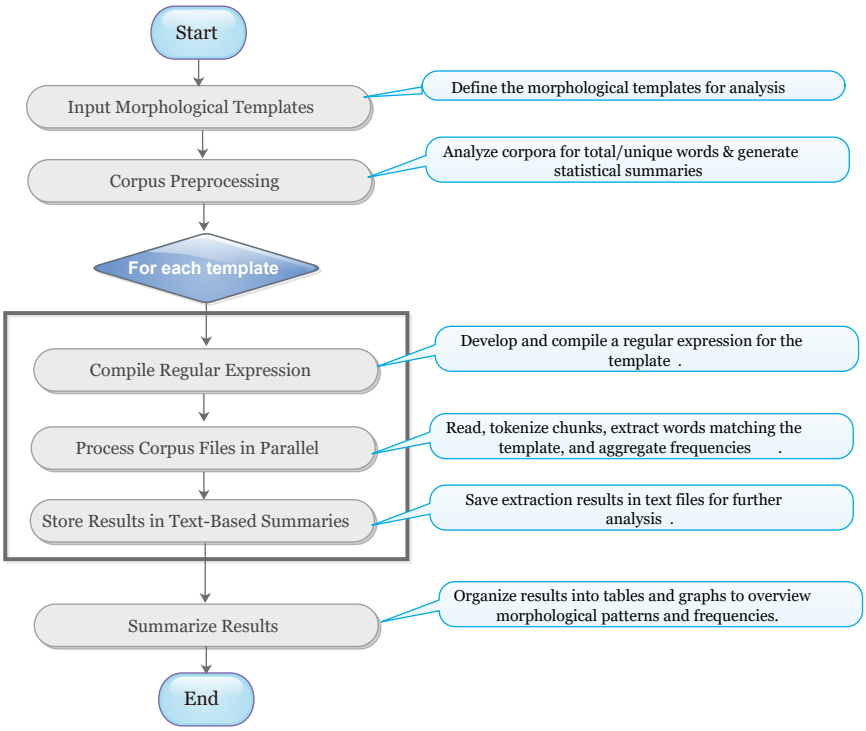

Fig. 6. A workflow: Structured steps for morphological pattern extraction and lexical resource development.

5 Discussion and Computational Applications

This framework serves as a computational model for Arabic morphological analysis, offering structured methodologies for pattern extraction, lexicon development, and corpus annotation. This section highlights the framework's integration in digital linguistic resources and NLP pipelines, distinguishing its technical contributions from broader linguistic findings.

5.1 Practical Applications

As a robust tool for Arabic morphological analysis, this framework plays a crucial role in computational linguistic research. Its structured methodology for pattern extraction and classification supports various applications across lexicon development, large-scale corpus annotation, and NLP workflows. Beyond academic research, this tool contributes to the enhancement of digital linguistic resources, providing practical benefits for both researchers and NLP practitioners.

5.1.1 Automated Morphological Pattern Dictionary Construction. The system systematically extracts and categorizes morphological structures using root-based and template-based approaches. By classifying both canonical and analogy-driven patterns such as formations derived from the root "ف-ع-ل" (f-'-l), it facilitates the automated creation of Arabic morphology databases, improving the structure and usability of digital lexical resources.

The following workflow (Figure 6) outlines the structured steps followed for morphological pattern extraction and lexical resource development.

5.1.2 Supporting Linguistic Research. Based on large-scale corpus analysis, this tool provides empirical validation for morphological productivity, distributional patterns, and syntactic constraints in the Arabic language. By identifying frequency and contextual usage of patterns such as "استفعال" (istifʿāl) across different genres, *Mizan* promotes more data-driven linguistic modeling, reducing annotation inconsistencies and enhancing computational morphological classification [13].

5.1.3 Integrating in NLP Pipelines. Structured outputs, namely, morphological templates, frequency distributions, and disambiguation data, can be seamlessly integrated in NLP pipelines. By refining part-of-speech tagging, lemma extraction, and morphological disambiguation, the tool improves downstream applications, such as machine translation, syntactic parsing, and entity recognition.

5.1.4 Educational Applications. The ability to generate structured summaries of high-frequency morphological patterns makes this system a valuable tool for teaching Arabic morphology. By facilitating the creation of educational materials and structured linguistic tasks, *Mizan* enhances language instruction for non-native learners and supports the development of curricula focused on productive Arabic morphological structures.

5.2 Enriching Digital Linguistic Resources

Beyond its role in education and NLP, this framework contributes significantly to the enhancement of digital linguistic infrastructures. Through large-scale corpus processing and lexicon development, the tool supports the creation of computational lexicons, annotated corpora, and search-enhanced linguistic datasets for broader NLP applications.

5.2.1 Annotation of Large Corpora. This system annotates large-scale corpora with detailed morphological markers, structuring these annotations into systematically organized datasets. These annotated resources serve as foundational building blocks for computational lexicons, enabling automated text analysis, morpho-syntactic parsing, and large-scale Arabic NLP applications [5, 35, 40, 44].

5.2.2 The Development of Computational Lexicons. By leveraging pattern extraction capabilities, the system facilitates the construction of structured digital lexicons, such as databases of Arabic verb patterns. Through the statistical analysis of word frequency and distribution, *Mizan* helps computational linguists refine existing lexical databases and develop domain-specific linguistic resources [4, 38].

Figure 7 illustrates how the proposed tool facilitates the creation of lexical resources through a structured, multi-step workflow. The process begins with defining morphological templates that reflect canonical Arabic patterns. These templates guide the pattern-based analysis of a large corpus, with extracted results stored and summarized according to pattern type. The final output consists of a structured report containing root-pattern pairings, frequency distributions, and part-of-speech classifications. This report serves as a foundational lexical resource, particularly for researchers focusing on derivational morphology.

For example, highly ambiguous patterns such as *tafāʿul* (تفاعل) or *infiʿāl* (انفعال) often yield both verbal and nominal forms (e.g., *tabādal*: تبادل = "exchange" vs. *tabādul*: تبادَلَ = "he exchanged"), which are distinguished through pattern frequency, root behavior, and syntactic context. Similarly, irregular forms like *istikāna*: استكانة (pattern: *istifʿāl*: استفعال) defy regular semantic predictions due to their lexicalization and require additional post-processing for accurate classification. These edge cases are handled using cumulative corpus statistics and optional user-defined disambiguation filters.

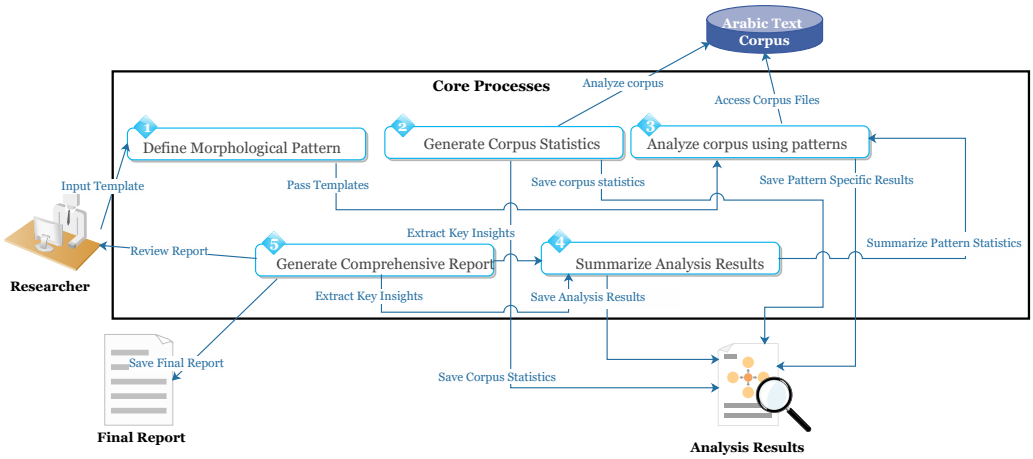

Fig. 7. Data flow: From morphological patterns to lexical resource creation.

5.2.3 Enhancing Searchability and Structured Querying of Linguistic Data. This approach enhances linguistic research by structuring morphological corpora into indexed query-ready datasets. It enables researchers to efficiently track morphological evolution, compare frequency distributions across genres, and analyze patterns in a structured manner. By transforming linguistic data into structured query-ready formats, the system enhances both human linguistic research and automated NLP workflows.

5.3 Tool Limitations and Future Insights

While this system demonstrates strong performance in Arabic morphological analysis, certain limitations persist due to the complexity of Arabic morphology and computational constraints. This section outlines specific enhancements to its underlying algorithms and architecture, focusing on resolving tool-specific limitations rather than analyzing broader linguistic challenges.

5.3.1 Improving Disambiguation for Highly Flexible Morphological Patterns. This approach effectively analyzes structured patterns such as "*istifʿāl* (استفعال)" and "*mafʿūl* (مفعول)" due to their well-defined morphology. However, highly ambiguous patterns like "*tafaʿʿul* (تفعّل)" and "*fāʿil* (فاعل)" present classification challenges. These forms serve multiple grammatical functions, including verb stems, active participles, and nominal derivatives. As a result, distinguishing their correct usage requires a context-aware approach rather than exclusive reliance on morphological templates.

To this end, some improvements can be suggested by allowing future versions to integrate probabilistic models trained on large annotated corpora. Combining deep learning embeddings with rule-based syntactic constraints can refine classification accuracy for morphologically overlapping patterns.

5.3.2 Handling Irregular and Phonologically Variable Forms: Beyond ambiguity, the tool encounters difficulties in recognizing irregular verb forms and phonologically shifting patterns, particularly within triliteral verb inflections. These forms exhibit variations that are not strictly rule-governed but rather phonetic adaptations, requiring a more dynamic analysis approach. Thus, some improvement can be proposed by implementing lexicon-augmented models that incorporate phonetic

variations from authoritative Arabic dictionaries in order to improve the system's ability to predict and normalize irregular forms.

5.3.3 Optimizing Scalability and Computational Efficiency. While this framework leverages parallel processing and caching for large-scale corpus analysis, handling highly inflected forms continues to pose computational challenges due to a high range of morphological variation and dataset complexity [18, 35, 43]. The current system employs parallel processing and caching, but both memory usage and processing speed can still be further optimized. Hence, some improvements can be suggested by exploring GPU-accelerated computation. Thus, more efficient indexing strategies could enhance processing speed while reducing computational overhead.

5.3.4 Key Development Priorities for Future Versions. To promote the tool's capabilities beyond its current architecture, future development efforts should prioritize the use of expanding lexical databases by integrating structured lexicons in real-world corpus data in order to refine word classifications. Moreover, the use of hybrid parsing models helps combine statistical NLP techniques and linguistically informed rules to balance accuracy and efficiency. In addition, improved contextual analysis plays a key role in leveraging transformer-based models (such as BERT [15] or AraBERT [6]) to enhance word sense disambiguation.

6 Results

This section presents the study's key findings regarding Mizan's performance in large-scale Arabic morphological analysis. The results demonstrate the tool's efficiency in processing extensive Arabic corpora, its accuracy in extracting morphological patterns, and its capacity to generate valuable linguistic insights. In addition, the study identifies key challenges such as morphological ambiguity, dialectal variation, and irregular word forms. The broader implications of these findings for computational linguistics, lexicography, and NLP applications are discussed, along with future directions for enhancing Mizan's adaptability and accuracy.

6.1 Efficiency and Scalability

The study confirmed that *Mizan* is capable of efficiently processing large-scale Arabic corpora, as evidenced by its successful handling of a dataset comprising 572 million words. Processing times varied depending on the morphological pattern analyzed, ranging from 14 minutes and 6 seconds for the pattern *mutafāʿil* (متفاعل) to 16 minutes and 25 seconds for *istifʿāl* (استفعال), reflecting the system's ability to manage a wide range of morphological structures. Performance was further enhanced through the use of parallel processing and caching, which significantly improved speed and memory management by optimizing resource utilization. Moreover, scalability was ensured by leveraging data partitioning and distributed computing frameworks such as Dask and Apache Spark, enabling the system to remain effective and responsive even when applied to significantly larger datasets.

6.2 Morphological Pattern Extraction and Linguistic Insights

The study demonstrated that longer and structurally well-defined morphological patterns, such as *istifʿāl* (استفعال) and *mutafāʿil* (متفاعل), achieved perfect extraction accuracy (F1 = 100%), a result attributed to their distinct and unambiguous morphological features. In contrast, shorter patterns such as *faʿīl* (فعيل) and *faʿūl* (فعول) exhibited lower precision, primarily due to their high morphological flexibility and overlap with other derivational forms, which introduced challenges in disambiguation. Despite this variation, precision and recall remained consistently strong across all patterns, with F1-scores ranging from 80.3% for *faʿīl* to 100% for *mutafāʿil* and *istifʿāl*, underscoring Mizan's Effectiveness in identifying and extracting standard morphological structures. Furthermore,

a distributional analysis of the extracted patterns revealed valuable linguistic insights into the relative productivity of different morphological templates in contemporary Arabic usage.

6.3 Challenges and Linguistic Variability

The evaluation revealed several linguistic and computational challenges that impact the accuracy and generalizability of large-scale morphological analysis. Morphological ambiguity, particularly prevalent in shorter and structurally flexible patterns, resulted in an increased rate of false positives. This underscores the need for more refined disambiguation techniques, as elaborated in Section 5.3. In addition, dialectal variation and the presence of loanwords posed classification difficulties, revealing the limitations of standard lexicons and highlighting the importance of incorporating dialect-specific lexical resources. Another challenge emerged from irregular morphological forms and the absence of diacritics, which introduced inconsistencies in verb and noun identification due to phonological variation and context sensitivity—indicating a need for context-aware modeling. Despite Mizan's optimizations in speed and resource management, the computational demands of processing highly inflected forms in massive corpora remain significant, calling for ongoing improvements in scalability and efficiency.

6.4 Broader Implications and Practical Applications

The findings of this study carry significant implications for various linguistic and computational domains. In lexicography, the ability to automatically extract root-pattern combinations provides a valuable foundation for constructing high-quality morphological dictionaries tailored to Arabic. From a linguistic research perspective, the structured data produced by Mizan enables comprehensive quantitative analyses of morphological phenomena such as pattern frequency, productivity, and distribution across large-scale corpora. In the field of NLP, the enhanced morphological representations contribute to improving core applications including machine translation, automatic text summarization, and semantic processing. Furthermore, the study's insights have pedagogical value, offering data-driven support for Arabic morphology instruction by identifying frequent and productive patterns relevant to curriculum development.

Finally, Mizan's underlying methodology demonstrates cross-domain adaptability, with potential for extension to domain-specific texts, such as legal, medical, or scientific corpora, thus broadening its applicability and linguistic coverage.

6.5 Future Directions

Building on the findings of this study, several avenues for future development have been identified. First, enhancing context-aware disambiguation remains a key priority, particularly through the integration of advanced contextual analysis techniques that can improve the precision of morphological pattern classification.

Second, incorporating machine learning approaches, both supervised and unsupervised, offers promising potential for boosting the accuracy of pattern recognition and expanding the adaptability of the system to linguistic variability. Third, the expansion of lexical resources, especially in the areas of dialectal Arabic and specialized domains, is essential for improving the tool's linguistic coverage and applicability.

Finally, continued efforts in scalability enhancement are necessary to ensure Mizan can process increasingly larger and more diverse corpora efficiently, making it suitable for real-world deployment in high-volume NLP tasks. These results reaffirm Mizan's effectiveness as a robust tool for large-scale Arabic morphological analysis. Its demonstrated capabilities in processing extensive corpora, extracting morphological patterns, and supporting linguistic inquiry position it as a valuable asset for NLP applications, lexicographic development, and computational linguistics. At the

same time, the challenges identified particularly in disambiguation and handling non-standard forms—underscore important directions for future research aimed at refining Arabic morphological modeling and enhancing contextual understanding [8, 48].

7 Conclusion

This study presents the design and development of *Mizan*, an innovative computational framework for Arabic morphological analysis in large-scale corpora. By systematically extracting and analyzing morphological patterns, the system provides detailed statistical insights into their distribution. Processing over 572 million words from the Arabic Wikipedia corpus demonstrated this tool's scalability and efficiency in handling complex linguistic data. The system's effectiveness is further validated by its performance metrics, achieving an F1 score of 100% for structurally distinct patterns (e.g., *mutafāʿil* (متفاعل), *istifʿāl* (استفعال)) and maintaining robust precision across various morphological templates (80.3%–100%). These results confirm Mizan's capability to extract standard morphological structures with high reliability, making it a valuable tool for large-scale Arabic text analysis.

A key strength of this approach lies in its ability to accurately extract complex patterns while addressing challenges posed by shorter patterns, such as "*faʿīl* (فعيل)" and " *faʿūl* (فعول)", where structural overlaps and the absence of diacritics contribute to ambiguities. These challenges reflect the inherent complexity of Arabic morphology and highlight areas for further refinement.

Beyond its technical performance, this framework contributes to computational linguistics by facilitating the systematic creation of morphological lexicons and annotated datasets. A core aspect of this work is the integration of structured morphological resources in various applications. The tool can enhance digital lexical resources by refining morphological dictionaries and enabling a structured classification of Arabic root-pattern formations, and improve NLP pipelines through the integration of structured lexicons to enhance lemmatization, part-of-speech tagging, and syntactic parsing. *Mizan* is also able to advance machine translation systems through more accurate Arabic morphological analysis and disambiguation and facilitate dialectal and domain-specific linguistic studies, using corpus-based pattern extraction and classification.

Future enhancements should focus on addressing identified limitations. Integrating context-sensitive algorithms and machine learning models would improve disambiguation, particularly for shorter and irregular patterns. Additionally, expanding the system's lexicon to include dialectal and domain-specific vocabularies would increase its adaptability to diverse linguistic contexts. Furthermore, broadening its scope to support specialized domains such as legal, scientific, and technical texts would enhance its utility beyond core linguistic research.

In conclusion, this study underscores the role of computational tools in tackling the complexities of Arabic morphology on a large scale. By refining its capabilities and broadening its applications, *Mizan* can serve as a valuable resource for Arabic computational linguistics, lexicography, education, technical documentation, and artificial intelligence-driven NLP. Insights from this study contribute to the ongoing evolution of computational linguistics and AI-driven NLP. By tackling the challenges of Arabic morphology, *Mizan* lays the groundwork for future advancements in scalable and adaptable morphological analysis, setting a benchmark for Arabic and other morphologically rich languages.

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